

Roll No.

TBT-401

B. TECH. (BIOTECH.) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

BIOPHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the role of hybridization in chemical bonding. (CO1)
(b) Describe briefly the forces that stabilize nucleic acid double helices. (CO1)
(c) Describe the X-ray diffraction technique in structural biology. (CO1)
2. (a) Discuss the Freely Jointed Chain (FJC) model. (CO2)
(b) Discuss the torsional angle and Ramachandran plot in Protein structure. (CO2)
(c) Describe the spontaneous and non-spontaneous process in terms of enthalpy, entropy and Gibbs free energy. (CO2)

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(2)

3. (a) Describe the mechanism of action potential in cell membrane. (CO3)
(b) Explain the resting membrane mechanism in Glial cell and Nerve cell. (CO3)
(c) Explain electrical equivalent circuit model for neuron. (CO3)
4. (a) Explain the working of cilia and flagella in Eukaryotes. (CO4)
(b) What are molecular motors ? Diagrammatically explain the characteristics of molecular motors. (CO4)
(c) How does the structure of microtubule relate to its function ? (CO4)
5. (a) Explain the new treatment technology for the treatment of peripheral artery disease. (CO5)
(b) What is the mechanobiology ? Give the significance of this field in relevance to the human health. (CO5)
(c) Write the characteristics and properties of nanomaterials and describe its application in biology. (CO5)

Roll No.

TBT-402

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

IMMUNOLOGY AND IMMUNOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are primary and secondary lymphoid organs ? With suitable diagram, explain the role of *Thymus* in acquired immune response.

(CO1)

P. T. O.

- (b) Describe in detail general characteristics of antigens. Also, discuss about antigenicity of 'Haptens' and role of adjuvants in enhancing their immunogenicity. (CO1)
- (c) What is Humoral and Cell mediated immune response ? Differentiate them using suitable examples. (CO2)
2. (a) Describe in detail general factors influencing the innate immune response. Also comment on anatomical and physiological barriers of innate immune response. (CO1)
- (b) What is phagocytosis ? Discuss the mechanism of killing pathogens via phagocytosis. (CO2)
- (c) What are subunit vaccines ? Describe the types of vaccines currently in use for immunization against COVID-19 infection. (CO3)
3. (a) What are complement proteins ? Explain classical pathway of their synthesis and mode of action. (CO2)
- (b) What are monoclonal antibodies ? Explain Hybridoma technology for their synthesis. (CO4)
- (c) What are MHC's ? Explain their different types and role in antigen presentation. (CO3)
4. (a) What is ELISA ? Explain the different types of ELISA used in immunological diagnostics. (CO4)
- (b) Comment on any *two* of the following : (CO4)
- (i) Haemagglutination

(3)

(ii) Immunofluorescence

(iii) RIA

- (c) Describe complement fixation test and its applications in immunodiagnostics of an antigen. (CO4)
5. (a) What are immune deficiencies ? Explain any *two* immune-deficiency disorders. (CO5)
- (b) Explain the types of transplant antigens. Describe the mechanism of transplant rejection by T-lymphocytes. (CO5)
- (c) What is hypersensitivity ? With suitable examples, explain Type-II hypersensitive immune response. (CO5)

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TBT-403

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

BIOANALYTICAL TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the role of bioanalytical techniques in various industries related to biotechnology. (CO1)
- (b) Differentiate between SEM and TEM. (CO1)
- (c) Explain the working of bright field microscopy. (CO1)
2. (a) Explain the working and applications of 2D SDSPAGE. (CO2)
- (b) Explain the working principle of centrifugation and its types. (CO2)
- (c) Enlist various applications of centrifugation and electrophoresis. (CO2)

P. T. O.

(2)

3. (a) Discuss the working principle of affinity chromatography. (CO3)
(b) Differentiate between working principle of paper and thin layer chromatography. (CO3)
(c) Write a short note on gas chromatography and its application. (CO3)
4. (a) Write a short note on working principle of spectrophotometry. (CO4)
(b) Discuss the working principle of Mass Spectroscopy and its various applications. (CO4)
(c) Write a short note on X-ray diffraction. (CO4)
5. (a) Enlist the properties of α , β and γ -rays and their uses. (CO5)
(b) Write a short note on radioactivity measurement method. (CO5)
(c) Write a short note on in-vivo and in-vitro labelling technique. (CO5)

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TBT-404

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

ENVIRONMENTAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the methods used for the measurement of environmental pollution. (CO1, CO2)

(b) Explain the elements responsible for causing acid rain and the ways by which you can reduce the impact of acid rain. (CO1, CO2)

P. T. O.

- (c) Give the criteria for the design of a bioreactor for the treatment of waste water. (CO1, CO2)
2. (a) Highlight the role of biocatalyst and its organic reaction mechanism. (CO3, CO4)
- (b) What are genetically engineered organisms ? Describe the ethical concerns imposed by the genetically engineered organisms on the environment. (CO3, CO4)
- (c) Explain the national standards for the disposal of waste. Also give the brief account on the up-flow anaerobic sludge blanket bioreactor. (CO3, CO4)
3. (a) Write different types of xenobiotic compounds. Give a detailed account on biodegradation of any aromatic xenobiotic compound. (CO4, CO5)
- (b) Comment on the technology developed by Indian scientists for the ecological restoration. (CO4, CO5)
- (c) Discuss the approaches in relevance to biotechnology for the protection of environment. (CO4, CO5)
4. (a) "Identification of effects of activities on the human health and environment and mitigates these impacts." Comment on the aforementioned statement. (CO1, CO5)
- (b) Highlight the role of biopesticides in integrated pest management. (CO1, CO5)

(3)

- (c) Explain consequences of water pollution. Discuss the need of water management. (CO1, CO5)
5. (a) Differentiate between Biological Oxygen Demand and Chemical Oxygen Demand. (CO1, CO2)
- (b) Comment on causes, mechanism of ozone depletion. Discuss various measures for preventions of ozone depletion. (CO1, CO2)
- (c) How does aquatic microorganisms help in reducing metal pollution? Explain with examples. (CO1, CO2)

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TBT-405(a)

B. TECH. (BIOTECH.) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
WASTE MANAGEMENT AND UPCYCLING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

- 06/7/2022
1. (a) Discuss the different types of air pollutants.
(b) What is Air Pollution ? Write an explanatory note on sources of air pollution.
(c) Discuss the different types of waste water treatment technologies in detail.
 2. (a) Discuss the innovative technologies for sustainable waste management.
(b) Write a note on types of composting.
(c) What is the role of waste up cycling in a social enterprise ?
 3. (a) Write a note on carbon footprinting.

P. T. O.

(2)

- (b) Discuss the role of civil engineering in the sustainable waste management.
 - (c) What is the landfill engineering ? Discuss about the management of landfill leachate.
4. (a) Write an explanatory note on the impact of different types waste on human health.
- (b) Discuss the recycling and recovery technologies of wastes.
 - (c) Write a detailed note on solid waste management.
5. (a) Discuss the various stages of waste water treatment.
- (b) Write short notes on pyrolysis and gasification.
 - (c) Define Biomedical waste. Discuss about the management methods of Biomedical waste.

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TBT-405(c)

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

BIOMATERIALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the common metals and polymers used in human implants ?

Explain with examples. (CO1)

(b) Explain the phase diagram of Iron-carbon with neat sketch. (CO1)

(c) Explain the following in brief : (CO1)

(i) Toughness

(ii) Elastic modulus

(iii) Surface hardness

(iv) Shear strength

P. T. O.

2. (a) What is calcination ? Discuss in brief with the necessary equation. (CO2)
- (b) What do you understand by solidification of metals ? How does it affect the crystal structure ? Explain in detail. (CO2)
- (c) Differentiate between stain steel and titanium biomaterial with examples. (CO2)
3. (a) Discuss various biopolymers obtained from plants, animals and agriculture products. (CO3)
- (b) What are the advantages of reinforcing fiber in polymer ? How come the fiber reinforcement polymer composite is better than filler reinforced polymer composites ? (CO3)
- (c) How can we reduce the bioreactivity of metal and ceramic biomaterial ? Explain in brief. (CO3)
4. (a) Discuss in detail the rolling process. How is it different from forming ? (CO4)
- (b) Explain in detail any three-surface finish process. (CO4)
- (c) Differentiate between extrusion and deep drawing. Write any *three*. (CO4)
5. (a) How are the hydrogels for bioimplants prepared ? Explain in detail. (CO5)
- (b) What are the common applications of hydrogels in human implant ? (CO5)
- (c) What is direct moulding process for making bioimplants ? Explain in brief. (CO5)

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TBT-406(a)

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

ALGORITHMS IN COMPUTATIONAL BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define molecular computing. How does molecular computing work ?
(b) Differentiate between biological and computer algorithms.
(c) What is DNA Computing ? Describe the scope and applications of DNA Computing.
2. (a) What is the difference between Binary Search and Linear Search ?
(b) Write the steps to sort an array $A[] = [40, 20, 25, 8, 45, 10]$ using insertion sort. Also provide pseudo code.
(c) What are the notations to calculate time and space complexity of an algorithm ?

P. T. O.

(2)

3. (a) What are the tradeoff between time and space complexity ?
(b) What algorithm does BLAST use ? Discuss the steps of BLAST algorithm.
(c) Write short notes on the following :
 - (i) Genetic Algorithm
 - (ii) Divide and Conquer Algorithm
 - (iii) Hamiltonian Graph
 - (iv) Dynamic Programming
4. (a) What are the advantages and limitations of hill climbing search algorithm ?
(b) What is pattern matching ? How Hash Table is used in Pattern Matching ?
(c) Describe the applications of Genetic Algorithm in Bioinformatics.
5. (a) Differentiate between Needleman-Wunsch and Smith-Waterman algorithm.
(b) Explain the Greedy algorithm and when it can be applied.
(c) What is hidden Markov model and its role in biology ?

(Btech Bio Tech)
End Semester Examination 2022
IV Semester
Mobile Application Development

Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Each question carries 10 marks.

Q1.

- a) Explain, how events are handled in JavaScript. Write an event handler program using function that takes a string which has lower- and upper-case letters as a parameter and converts upper-case letters to lower-case, and lower-case letters to upper case (CO1)
- b) Explain how to use Cordova and the Ionic Framework to create a web-based mobile application. (CO2)
- c) Explain the process of creating a native application. What are the gain and downside with native application? (CO1)

Q2.

- a) Write the code in java script convert given string binary to decimal use prompt for input and alert for output. (CO1)
- b) In Android, how do you define an activity? What layouts are available on Android? (CO2)
- c) In Android, what is a broadcast receiver? What exactly is the Android ecosystem? (CO2)

Q3.

- a) Write the code in java script to create a html page to display a new image & text when the mouse comes over the existing content in the page. (CO1)
- b) What is the meaning of intent in Android app development? In the creation of mobile applications, explain types of intent. (CO3)
- c) Compare and contrast mobile app development with conventional app development. With appropriate instances. (CO3)

Q4.

- a) What is the difference between static and dynamic web pages? Which technologies we have discussed in our course are needed to design and develop them? Please explain their objectives and how these interact with each other to create a website? (CO1)
- b) Describe the architecture of android operation system with suitable diagram. (CO2)
- c) What are the key design elements of an Android application? (CO1)

Q5.

- a) What are various ways to include CSS and JavaScript in your web pages? Explain all of them by taking a simple example like taking array of numbers form user, find reverse of each number and stored in separate array by using a double click on button. (CO1)
- b) Explain the file structure of Android Studio, which is used to build mobile applications. (CO2)
- c) Using a neat sketch, describe the life cycle of an android activity. (CO3)